

Federated HiT-NeXt for Multilabel ECG Abnormality Classification under Cross-Dataset Heterogeneity

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Abstract

Electrocardiograms (ECGs) are inexpensive, non-invasive exams widely used for cardiovascular assessment, but robust deep learning models for ECG interpretation typically require large and diverse datasets. In clinical settings, such datasets are often distributed across institutions and are difficult to centralize because ECG traces and patient metadata are sensitive health data. This project studies federated learning (FL) as a privacy-aware training strategy for multilabel ECG abnormality classification across three public 12-lead ECG datasets: CODE-15, PTB-XL, and Chapman-Shaoxing. Each dataset is treated as a simulated institutional client with its own label prevalence, demographic structure, and multilabel burden. We use HiT-NeXt, a hybrid hierarchical ECG architecture combining convolutional patch merging and windowed transformer blocks, and evaluate supervised local training, balanced FL, and unbalanced FL. The experiments address four research questions: whether FL benefits low-data clients, whether this benefit persists with more local data, whether high-resource clients help low-resource clients, and whether collaboration harms high-resource clients. Results show that balanced FL clearly improves performance in the 1k-data regime, especially in macro F1 and cross-domain generalization. With 5k and 10k examples per client, local in-domain gains diminish and individual training often matches or exceeds FL locally, but FL remains beneficial relative to foreign-model transfer. In unbalanced FL, high-resource clients strongly improve the low-resource client, increasing low-resource macro F1 by 0.294 on average over individual 1k training and by 0.164 over balanced 1k FL. This comes with a small average macro-F1 cost of 0.027 for the high-resource clients. Overall, the results support FL mainly as a mechanism for low-resource and cross-domain robustness rather than as a universal replacement for strong in-domain training.

1 Introduction

Cardiovascular diseases remain one of the leading global causes of mortality, and the electrocardiogram (ECG) is among the most common tools for evaluating cardiac rhythm, conduction, and morphology in routine care. A standard 12-lead ECG records the electrical activity of the heart over time from multiple spatial projections. It is inexpensive, non-invasive, and clinically informative, supporting the detection of conduction abnormalities, rhythm disturbances, rate abnormalities, ischemic patterns, and broader cardiovascular risk. These properties have made ECGs a natural target for automated interpretation with machine learning and deep learning.

Deep learning has substantially changed automated ECG analysis. Earlier ECG systems often relied on hand-designed features such as QRS morphology, RR intervals, wavelet transforms, or other signal-processing descriptors followed by classical classifiers. Modern methods instead learn representations directly from raw ECG traces. Large-scale convolutional neural networks have reached cardiologist-level performance for arrhythmia detection and 12-lead abnormality classification [2, 8]. Public and semi-public ECG resources such as PTB-XL [13], CODE/CODE-15 [8, 9], and Chapman-Shaoxing [15] have further enabled reproducible studies on ECG

classification. At the same time, ECGs are not simply generic time series. They contain clinically meaningful structure at multiple temporal scales: local wave morphology, intra-beat intervals, beat-to-beat distances, and global rhythm patterns. This motivates architectures that can combine local signal modeling with longer-range temporal context.

HiT-NeXt was proposed precisely for this type of hierarchical ECG representation learning [1]. It combines convolutional modules inspired by ConvNeXt-style design principles [6, 14] with transformer blocks based on local attention windows, following the broader insight that attention can be made more efficient and structured through local windows and shifted windows [5, 12]. In ECG analysis, this design is clinically motivated: conduction disorders such as first-degree atrioventricular block depend on intervals such as the PR interval, bundle branch blocks depend on local QRS morphology, and rhythm disorders such as atrial fibrillation depend on longer-range irregularity. HiT-NeXt therefore provides a strong backbone for studying collaborative ECG training under heterogeneous data distributions.

Despite progress in ECG modeling, data availability remains a central obstacle. Strong supervised models often require many labeled ECGs, but real clinical data are distributed across hospitals, health systems, devices, and populations. Centralizing such data is difficult because of legal, ethical, institutional, and technical constraints. Even when data can be shared, differences in patient populations, acquisition devices, label policies, prevalence, and clinical workflows can make pooled training nontrivial. Federated learning (FL) is a natural response to this setting. In FL, multiple clients collaboratively train a model while keeping raw data local; only model parameters or updates are communicated to a central server [3, 7]. This paradigm is especially relevant for healthcare, where data are sensitive and distributed across institutions [10, 11]. However, FL also faces statistical heterogeneity: clients may have non-identically distributed data, different label prevalence, different demographic structure, and different sample sizes [3, 4].

This project investigates FL for ECG classification in a controlled cross-dataset setting. The three public datasets, CODE-15, PTB-XL, and Chapman-Shaoxing, are treated as simulated institutional clients. This setup is not a production deployment of privacy-preserving medical AI; rather, it is an experimental proxy for the institutional silo setting. It allows us to ask whether collaborative training helps when clients have little local data, whether gains persist as local data increase, and how the benefits and costs are distributed between low-resource and high-resource clients. The target task is multilabel classification over six ECG abnormalities: first-degree atrioventricular block (1dAVb), right bundle branch block (RBBB), left bundle branch block (LBBB), sinus bradycardia (SB), atrial fibrillation (AF), and sinus tachycardia (ST).

The guiding research questions are:

1. **RQ1: Does FL benefit scenarios with few local examples?** We evaluate whether balanced FL improves performance when each client has only 1k local training ECGs.
2. **RQ2: Does the FL gain persist with more local data?** We test whether the advantage remains when each client has 5k or 10k local training ECGs.
3. **RQ3: Do larger clients help a low-resource client?** We evaluate an unbalanced scenario in which one client has 1k examples and the other two clients have 10k examples.
4. **RQ4: Does collaboration harm high-resource clients?** We quantify whether participating in unbalanced FL reduces the performance of the 10k clients.

The main finding is that the value of FL depends strongly on the data regime. Balanced FL is most useful in the 1k regime, where it improves all clients in macro F1 relative to individual training. With 5k and 10k examples, the local in-domain advantage becomes weaker and sometimes negative, but the federated model remains better than using foreign models trained on other datasets. In unbalanced FL, high-resource clients substantially benefit the low-resource client, while the average cost to high-resource clients is small. Thus, in this setting, FL is

best interpreted as a mechanism for low-resource support and cross-domain robustness under heterogeneous ECG distributions.

2 Experimental Setup

2.1 Task formulation

Each ECG is represented as a 12-lead signal and assigned a six-dimensional binary target vector:

$$y_i \in \{0, 1\}^6,$$

where the target dimensions correspond to 1dAVb, RBBB, LBBB, SB, AF, and ST. Because each exam can have zero, one, or multiple positive abnormalities, the problem is formulated as multilabel classification. The model outputs one logit per abnormality:

$$f_{\theta}(x_i) \in \mathbb{R}^6.$$

Training minimizes binary cross-entropy with logits:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^6 [y_{ik} \log \sigma(f_{\theta,k}(x_i)) + (1 - y_{ik}) \log(1 - \sigma(f_{\theta,k}(x_i)))].$$

All reported metrics are macro averages over the six target abnormalities. This is important because the task is highly imbalanced, especially in CODE-15, where most ECGs have no positive target. Macro averaging gives each target equal weight and avoids allowing common negative labels or common abnormalities to dominate the evaluation.

2.2 Datasets as federated clients

The three datasets are treated as three separate FL clients:

- **CODE-15**: a Brazilian public subset of the CODE ECG cohort, derived from 12-lead ECGs collected by the Telehealth Network of Minas Gerais.
- **PTB-XL**: a large European public 12-lead ECG dataset with structured diagnostic annotations.
- **Chapman-Shaoxing**: a public Chinese 12-lead ECG dataset for arrhythmia research.

This client construction deliberately creates a heterogeneous federation. The clients differ in sample size, positive-label density, class prevalence, and multilabel structure. Therefore, the experiment is closer to a cross-silo FL setting than to an IID partition of a single dataset.

Table 1: Dataset overview after harmonizing targets to the six multilabel ECG abnormalities.

Dataset	Records	Positive labels	Mean labels/record	Records with any target	Any target %
CODE-15	345779	41636	0.120	37775	10.92
PTB-XL	21799	6041	0.277	5334	24.47
Chapman-Shaoxing	44539	27359	0.614	25822	57.98

The dataset summary in Table 1 shows that the clients differ substantially before model training begins. CODE-15 is by far the largest dataset, but it is also the sparsest in target abnormalities: only 10.92% of records contain at least one positive target. PTB-XL is smaller but denser, with 24.47% of records containing at least one target and a mean of 0.277 positive labels per record. Chapman-Shaoxing is the densest client: 57.98% of records contain at least one target and the mean number of positive labels per record is 0.614, more than five times the CODE-15 value. These differences imply that the federated setting is not only heterogeneous in dataset size, but also in the effective supervision signal available to each client.

2.3 Statistical heterogeneity

The prevalence of the six target abnormalities varies strongly across clients, as shown in Table 2. This is central to the interpretation of the FL results because a model trained on one client sees a different empirical task distribution from a model trained on another client.

Table 2: Class prevalence by dataset.

Dataset	Class	Positive records	Prevalence
CODE-15	1dAVb	5716	1.65%
	RBBB	9672	2.80%
	LBBB	6026	1.74%
	SB	5605	1.62%
	AF	7033	2.03%
	ST	7584	2.19%
PTB-XL	1dAVb	793	3.64%
	RBBB	1658	7.61%
	LBBB	613	2.81%
	SB	637	2.92%
	AF	1514	6.95%
	ST	826	3.79%
Chapman-Shaoxing	1dAVb	1137	2.55%
	RBBB	648	1.45%
	LBBB	239	0.54%
	SB	16522	37.10%
	AF	1779	3.99%
	ST	7034	15.79%

The most striking difference is in rate abnormalities. Sinus bradycardia appears in only 1.62% of CODE-15 and 2.92% of PTB-XL but in 37.10% of Chapman-Shaoxing. Sinus tachycardia appears in 2.19% of CODE-15 and 3.79% of PTB-XL but in 15.79% of Chapman-Shaoxing. Conversely, RBBB and AF are more prevalent in PTB-XL than in Chapman-Shaoxing. These differences imply that cross-domain transfer is not simply a question of signal morphology; it is also affected by label priors, decision thresholds, and the precision-recall tradeoff induced by each dataset.

Table 3: Largest pairwise prevalence differences between clients.

Dataset A	Dataset B	Class	Prev. A	Prev. B	Abs. diff.
CODE-15	Chapman-Shaoxing	SB	1.62%	37.10%	35.47%
PTB-XL	Chapman-Shaoxing	SB	2.92%	37.10%	34.17%
CODE-15	Chapman-Shaoxing	ST	2.19%	15.79%	13.60%
PTB-XL	Chapman-Shaoxing	ST	3.79%	15.79%	12.00%
PTB-XL	Chapman-Shaoxing	RBBB	7.61%	1.45%	6.15%
CODE-15	PTB-XL	AF	2.03%	6.95%	4.91%
CODE-15	PTB-XL	RBBB	2.80%	7.61%	4.81%
PTB-XL	Chapman-Shaoxing	AF	6.95%	3.99%	2.95%
PTB-XL	Chapman-Shaoxing	LBBB	2.81%	0.54%	2.28%
CODE-15	PTB-XL	1dAVb	1.65%	3.64%	1.98%

The multilabel burden also differs substantially. In CODE-15, 89.08% of ECGs have no

positive target. In PTB-XL, the corresponding fraction is 75.53%. In Chapman-Shaoxing, only 42.02% of ECGs have no positive target, while 54.65% have exactly one target. Therefore, Chapman-Shaoxing is not only denser in positives but also has a different negative-to-positive balance. This affects loss gradients, threshold selection, and macro metrics.

Table 4: Multilabel burden: number of positive target labels per record.

Dataset	Positive labels/record	Records	Fraction
CODE-15	0	308004	89.08%
	1	34104	9.86%
	2	3486	1.01%
	3	180	0.05%
	4	5	0.00%
PTB-XL	0	16465	75.53%
	1	4652	21.34%
	2	657	3.01%
	3	25	0.11%
Chapman-Shaoxing	0	18717	42.02%
	1	24339	54.65%
	2	1432	3.22%
	3	48	0.11%
	4	3	0.01%

Demographic structure differs as well. CODE-15 has mean age 53.24 years, PTB-XL has mean age 62.77 years, and Chapman-Shaoxing has mean age 58.31 years. PTB-XL is approximately balanced by sex, while Chapman-Shaoxing has a larger male fraction. These demographic differences are not used as target labels in this project, but they are relevant because ECG morphology and abnormality prevalence vary with age and sex.

Table 5: Age and sex summaries.

Dataset	Age N	Missing	Mean	Std.	Median	IQR
CODE-15	345779	0	53.240	19.695	54.000	37–70
PTB-XL	21799	0	62.769	32.309	62.000	50–72
Chapman-Shaoxing	44484	55	58.306	19.666	62.000	48–72

Dataset	Sex	Count	Fraction
PTB-XL	female	11354	52.08%
PTB-XL	male	10445	47.92%
Chapman-Shaoxing	female	19334	43.41%
Chapman-Shaoxing	male	25183	56.54%
Chapman-Shaoxing	missing	22	0.05%

2.4 Model: HiT-NeXt

All experiments use the HiT-NeXt ECG architecture [1]. HiT-NeXt is a hybrid hierarchical model that combines convolutional and transformer components. It was designed around the observation that ECG interpretation requires both local morphological analysis and longer-range temporal context. Local ECG details include wave shape, QRS morphology, and short intervals; global context includes rhythm regularity and rate over several beats.

The implementation used in this project follows a four-stage hierarchical configuration. The input ECG has shape (L, C) , where L is the number of temporal samples and $C = 12$ is the

number of leads. At each stage, convolutional patch merging reduces temporal resolution and increases the channel dimension. The model uses hidden dimensions [96, 192, 384, 768], layer counts [4, 4, 12, 4], attention heads [4, 8, 16, 32], downsampling factor 4, window size 10, and expansion factor 4. The classifier head outputs six logits, one per target abnormality.

The model has four important components:

1. **Convolutional patch merging.** Convolutional blocks aggregate local waveform information and reduce temporal resolution. This is useful for ECG signals because local morphology, such as QRS shape, P-wave structure, and sharp transitions, carries diagnostic information.
2. **Windowed transformer attention.** Transformer blocks use restricted local attention windows rather than full global attention. This keeps attention focused on local temporal neighborhoods and reduces the risk that global attention obscures fine-grained ECG patterns.
3. **Hierarchical multi-stage processing.** By progressively reducing resolution, later stages operate over larger effective temporal contexts. This allows the model to move from wave-level patterns to beat-level and rhythm-level representations.
4. **Relative/contextual positional information.** The implementation uses a combined relative positional encoding configuration, incorporating both distance-based relative position information and contextual positional encoding. This is important because several ECG abnormalities are defined by timing relationships, such as prolonged PR interval in 1dAVb or altered RR intervals in bradycardia and tachycardia.

This architecture is suitable for the federated experiment because it is expressive enough to model ECG abnormalities while being shared across clients without changing client-specific architecture. Therefore, differences in performance can be interpreted mainly through data regime, data heterogeneity, and training protocol rather than through changes in the backbone.

2.5 Training protocols

Three training protocols are evaluated.

Individual supervised training. A separate model is trained independently on each dataset. Experiments are run with 1k, 5k, and 10k local training examples. Each trained model is evaluated both in-domain and cross-domain on the other datasets. This protocol measures how well each individual client can perform alone and how well its learned representation transfers to other clients.

Balanced federated learning. All three clients participate in a federated training run with the same training-data budget per client. The evaluated regimes are 1k, 5k, and 10k examples per client. The server initializes a global model, sends it to selected clients, each client trains locally, and the server aggregates the returned model states using FedAvg-style model averaging [7]. In the balanced setting, because each client has the same local budget, weighted and uniform aggregation are effectively aligned.

Unbalanced federated learning. One client is restricted to 1k examples while the other two clients use 10k examples each. This simulates a low-resource institution collaborating with larger institutions. Each dataset is used once as the low-resource client, producing three unbalanced scenarios. This protocol directly addresses whether larger clients help a small client and whether the small client harms the larger clients.

2.6 Federated optimization

Let K be the number of participating clients in a given round. At round t , the server sends the global parameters θ_t to each selected client. Client k performs local training on its private dataset and returns updated parameters $\theta_{t+1}^{(k)}$. The server aggregates the local models as

$$\theta_{t+1} = \sum_{k=1}^K \frac{n_k}{\sum_{j=1}^K n_j} \theta_{t+1}^{(k)},$$

where n_k is the number of training examples used by client k . This is the standard FedAvg principle. The project implementation also supports uniform aggregation, but the default configuration uses weighted aggregation. In balanced regimes, this distinction is negligible because all clients use the same local sample count. In unbalanced regimes, weighted aggregation gives greater influence to the two 10k clients, which is relevant when interpreting the low-resource gains.

2.7 Evaluation protocol

For each model, the main metrics are accuracy, precision, recall, and F1, macro-averaged over the six target abnormalities. The central metric in the discussion is macro F1 because the task is multilabel and imbalanced. Cross-domain evaluations use thresholds selected on the target validation set. Therefore, cross-domain results represent *target-calibrated transfer*, not calibration-free transfer. This choice isolates representation transfer and classification ability from the additional problem of threshold calibration across domains.

For individual models, the generalization gap is defined as:

$$\text{Gap} = \text{MeanCrossDomain} - \text{InDomain}.$$

Positive values indicate that the mean cross-domain score is higher than the in-domain score, while negative values indicate worse transfer outside the training dataset. For FL, two deltas are reported:

$$\Delta_{\text{target}} = \text{FL} - \text{IndividualInDomain},$$

and

$$\Delta_{\text{foreign}} = \text{FL} - \text{ForeignModelMean}.$$

The first measures whether FL improves the target client compared with its own local model. The second measures whether FL improves over relying on models trained on other institutions.

3 Results and Discussion

3.1 Individual training and cross-domain transfer

Table 6 summarizes the in-domain and mean cross-domain macro F1 for individually trained models. Several patterns are visible. First, increasing local data generally improves in-domain performance. For CODE-15, in-domain F1 increases from 0.103 in the 1k regime to 0.663 in the 10k regime. For PTB-XL, it increases from 0.276 to 0.778. For Chapman-Shaoxing, it increases from 0.376 to 0.565. Second, the generalization behavior differs by training dataset. CODE-15 transfers relatively well on average, especially at 5k, where its in-domain F1 is 0.507 and mean cross-domain F1 is 0.502. PTB-XL obtains the strongest in-domain F1 but has a large cross-domain gap, especially at 5k and 10k. Chapman-Shaoxing also shows substantial generalization gaps, reflecting its distinct label distribution, especially the high prevalence of SB and ST.

Table 6: Individual-model macro-F1 generalization.

Training dataset/regime	In-domain F1	Mean cross-domain F1	Gap
CODE-15 1k	0.103	0.163	0.060
CODE-15 5k	0.507	0.502	-0.005
CODE-15 10k	0.663	0.573	-0.091
PTB-XL 1k	0.276	0.183	-0.093
PTB-XL 5k	0.748	0.472	-0.276
PTB-XL 10k	0.778	0.453	-0.325
Chapman 1k	0.376	0.178	-0.197
Chapman 5k	0.503	0.280	-0.223
Chapman 10k	0.565	0.407	-0.158

This first result already confirms that the clients are not interchangeable. A model trained on PTB-XL can achieve high PTB-XL performance but does not transfer equally well to CODE-15 and Chapman-Shaoxing. A model trained on Chapman-Shaoxing sees many positives and learns under a very different label prior, which can improve recall but reduce precision when transferred to sparser datasets. Thus, cross-domain transfer depends not only on sample size but also on how the source client’s label and demographic distributions align with the target.

3.2 RQ1: Does FL benefit scenarios with few local examples?

The answer is yes. In the 1k regime, balanced FL improves macro F1 for all three clients relative to individual in-domain training. Table 7 shows that CODE-15 improves from 0.103 to 0.315, PTB-XL from 0.276 to 0.412, and Chapman-Shaoxing from 0.376 to 0.417. The gains are largest for CODE-15 and PTB-XL in absolute terms, but all clients benefit.

Table 7: Balanced FL macro F1 compared with individual and foreign-model baselines.

Regime	Target	Individual	Foreign mean	Balanced FL	Δ_{target}	Δ_{foreign}
1k	CODE-15	0.103	0.103	0.315	0.212	0.212
1k	PTB-XL	0.276	0.184	0.412	0.135	0.227
1k	Chapman	0.376	0.237	0.417	0.041	0.180
5k	CODE-15	0.507	0.281	0.409	-0.098	0.127
5k	PTB-XL	0.748	0.489	0.630	-0.118	0.142
5k	Chapman	0.503	0.484	0.558	0.056	0.074
10k	CODE-15	0.663	0.341	0.538	-0.126	0.197
10k	PTB-XL	0.778	0.581	0.721	-0.057	0.141
10k	Chapman	0.565	0.511	0.617	0.052	0.105

The 1k results support the interpretation that FL is especially valuable when local supervision is scarce. In this regime, each local model has limited exposure to positive examples, especially for rare abnormalities. The federated model effectively sees a broader set of ECG patterns and label configurations while still keeping data local. This improves not only in-domain performance but also generalization relative to the average foreign model. The Δ_{foreign} values are all strongly positive in the 1k regime: +0.212 for CODE-15, +0.227 for PTB-XL, and +0.180 for Chapman-Shaoxing.

The improvement is not uniform across metrics. In the 1k regime, FL increases precision substantially for all clients and improves recall for PTB-XL and Chapman-Shaoxing. For CODE-15, however, FL recall is lower than the foreign-model mean but still higher than the

individual CODE-15 model. This reflects the tension between precision and recall in sparse datasets. CODE-15 contains many negatives, so a model that becomes more conservative can improve precision and F1 while not maximizing recall.

3.3 RQ2: Does the gain persist with more local data?

The answer is mixed. The local in-domain advantage of FL decreases as each client obtains more data. In the 5k regime, FL is worse than individual training on CODE-15 and PTB-XL in macro F1, but better on Chapman-Shaoxing. In the 10k regime, the same pattern remains: FL is lower than individual training on CODE-15 and PTB-XL but higher on Chapman-Shaoxing.

This suggests that once a client has enough local examples, a specialized in-domain model can exploit local label prevalence and domain-specific patterns more effectively than a single global model. The effect is especially clear for PTB-XL. The individual PTB-XL model reaches F1 0.748 at 5k and 0.778 at 10k, while balanced FL reaches 0.630 and 0.721. PTB-XL appears to benefit from specialization because it has a different prevalence profile, especially for RBBB and AF.

However, FL remains beneficial relative to foreign-model transfer. Even in the 5k and 10k regimes, Δ_{foreign} is positive for every target dataset. This means that a federated global model is consistently better than simply using models trained on other datasets. Therefore, the role of FL changes with data size. At 1k, it is both a local performance enhancer and a generalization enhancer. At 5k and 10k, it becomes more clearly a generalization and collaboration mechanism, while the best in-domain model may be local and specialized.

Table 8: Average balanced FL impact across data regimes by target dataset.

Target	Metric	Individual	Foreign	Balanced FL	Δ_{target}	Δ_{foreign}
Chapman	accuracy	0.899	0.837	0.875	-0.024	0.038
Chapman	precision	0.451	0.361	0.468	0.017	0.107
Chapman	recall	0.656	0.647	0.770	0.114	0.123
Chapman	f1	0.481	0.411	0.531	0.050	0.120
CODE-15	accuracy	0.953	0.862	0.974	0.021	0.112
CODE-15	precision	0.391	0.213	0.453	0.062	0.240
CODE-15	recall	0.565	0.532	0.469	-0.097	-0.063
CODE-15	f1	0.424	0.242	0.420	-0.004	0.179
PTB-XL	accuracy	0.905	0.849	0.949	0.044	0.101
PTB-XL	precision	0.571	0.383	0.523	-0.048	0.140
PTB-XL	recall	0.714	0.610	0.728	0.014	0.118
PTB-XL	f1	0.601	0.418	0.588	-0.013	0.170

Table 8 shows that, averaged across regimes, balanced FL improves macro F1 on Chapman-Shaoxing by 0.050, is essentially tied with CODE-15 (-0.004), and is slightly below PTB-XL (-0.013). However, compared with foreign models, FL improves macro F1 by 0.120 on Chapman-Shaoxing, 0.179 on CODE-15, and 0.170 on PTB-XL. This supports the slide-level conclusion: the local gain decreases with more data, but the global model still provides better cross-dataset robustness.

3.4 RQ3: Do larger clients help a low-resource client?

The answer is clearly yes. In the unbalanced setting, one client has 1k examples and the other two clients have 10k each. Table 9 shows that the low-resource client improves in every scenario. For CODE-15 as the 1k client, F1 improves from 0.103 with individual training to 0.415 with unbalanced FL. For PTB-XL, it improves from 0.276 to 0.636. For Chapman-Shaoxing, it improves from 0.376 to 0.585.

Table 9: Impact of unbalanced FL on the low-resource 1k client.

1k client	Metric	Individual 1k	Balanced FL 1k	Unbalanced FL	vs individual	vs balanced
CODE-15	accuracy	0.918	0.963	0.975	0.057	0.012
CODE-15	precision	0.103	0.344	0.485	0.381	0.140
CODE-15	recall	0.219	0.324	0.479	0.259	0.155
CODE-15	f1	0.103	0.315	0.415	0.312	0.100
PTB-XL	accuracy	0.766	0.916	0.963	0.197	0.046
PTB-XL	precision	0.248	0.361	0.579	0.331	0.218
PTB-XL	recall	0.533	0.568	0.777	0.244	0.210
PTB-XL	f1	0.276	0.412	0.636	0.359	0.224
Chapman	accuracy	0.810	0.730	0.944	0.134	0.214
Chapman	precision	0.339	0.357	0.542	0.203	0.185
Chapman	recall	0.619	0.767	0.793	0.174	0.026
Chapman	f1	0.376	0.417	0.585	0.209	0.168

The gains over balanced 1k FL are also positive. This is important because it shows that the benefit is not only due to federation itself, but also due to the presence of higher-resource collaborators. The average unbalanced low-resource improvement is +0.294 macro F1 relative to individual 1k training and +0.164 relative to balanced 1k FL. Therefore, high-resource clients transfer useful information to the low-resource client, even under substantial cross-dataset heterogeneity.

The PTB-XL low-resource case benefits most in macro F1 (+0.359 over individual and +0.224 over balanced FL). This likely reflects a favorable combination of effects: PTB-XL has moderate label density, and the two larger clients expose the global model to many more positives and broader ECG variability. CODE-15 also benefits strongly, especially in precision, increasing from 0.103 to 0.485. Chapman-Shaoxing benefits less in absolute F1 over individual training, but still improves substantially from 0.376 to 0.585. The fact that all three clients improve supports the use of FL for institutions with scarce labeled ECGs.

3.5 RQ4: Does collaboration harm high-resource clients?

The answer is yes, but only modestly on average, and not uniformly. Table 10 summarizes the average effect on each high-resource dataset across the unbalanced scenarios in which that dataset appears as a 10k client. CODE-15 suffers the largest average macro-F1 reduction, from 0.663 to 0.559 (-0.105). PTB-XL decreases from 0.778 to 0.735 (-0.043). Chapman-Shaoxing improves from 0.565 to 0.633 (+0.068). Across all high-resource clients and scenarios, the mean macro-F1 impact is -0.027.

The high-resource cost is small compared with the low-resource benefit. Low-resource clients gain +0.294 macro F1 on average over individual training, while high-resource clients lose only -0.027 macro F1 on average. This asymmetry is the main practical argument in favor of unbalanced FL: the low-resource institution receives a large benefit, and the high-resource institutions pay a limited average cost.

However, the cost is not negligible for every dataset. CODE-15 loses the most as a high-resource client. This may reflect its sparse positive-label distribution. When CODE-15 collabo-

Table 10: Average impact of unbalanced FL by high-resource dataset.

10k dataset	Metric	Individual 10k	Unbalanced FL	FL delta
Chapman	accuracy	0.964	0.961	-0.003
Chapman	precision	0.549	0.593	0.044
Chapman	recall	0.676	0.753	0.076
Chapman	f1	0.565	0.633	0.068
CODE-15	accuracy	0.985	0.976	-0.010
CODE-15	precision	0.602	0.504	-0.098
CODE-15	recall	0.780	0.707	-0.073
CODE-15	f1	0.663	0.559	-0.105
PTB-XL	accuracy	0.978	0.971	-0.006
PTB-XL	precision	0.759	0.687	-0.072
PTB-XL	recall	0.805	0.838	0.033
PTB-XL	f1	0.778	0.735	-0.043

rates with denser datasets, especially Chapman-Shaoxing, the global model may shift toward higher positive rates or different decision boundaries. Even with target-calibrated thresholds, this can reduce precision and F1 on CODE-15. Chapman-Shaoxing, in contrast, improves as a high-resource client, suggesting that exposure to CODE-15 and PTB-XL may regularize its dense and skewed label distribution.

Table 11 shows that the effect on high-resource clients also depends on which dataset is low-resource. When CODE-15 is the 1k client, the two high-resource clients slightly improve in average F1 (+0.007), driven by a large recall increase. When Chapman-Shaoxing or PTB-XL is the low-resource client, the high-resource clients experience modest average F1 decreases. This indicates that the direction of heterogeneity matters: the same FL algorithm can have different tradeoffs depending on which institution is small and which ones dominate the federation.

3.6 Final aggregate comparison

Table 12 aggregates the main differences across all experiments. Individual cross-domain transfer is worse than in-domain training by -0.145 macro F1 on average. Balanced FL is approximately tied with individual in-domain training (+0.011 macro F1 overall) but strongly better than foreign-model transfer (+0.156). Unbalanced FL gives the low-resource client a large gain (+0.294 over individual 1k and +0.164 over balanced 1k), while the high-resource clients experience a small average decrease (-0.027).

The precision and recall columns provide additional insight. Balanced FL improves precision over foreign models by +0.162 and recall by +0.060, yielding a large F1 gain. In unbalanced FL, the low-resource improvement is not merely a recall effect: precision improves by +0.305 and recall by +0.226 relative to individual training. This means the low-resource client benefits both from detecting more positives and from reducing false positives. For high-resource clients, precision decreases on average (-0.042), while recall slightly increases (+0.012). This suggests that the global unbalanced model tends to become somewhat more sensitive, but at a cost in precision.

Table 11: Average impact on the two high-resource clients grouped by the low-resource client.

1k dataset	Metric	Individual 10k	Unbalanced FL	FL delta
Chapman	accuracy	0.982	0.979	-0.002
Chapman	precision	0.680	0.636	-0.045
Chapman	recall	0.792	0.746	-0.046
Chapman	f1	0.721	0.674	-0.047
CODE-15	accuracy	0.971	0.962	-0.009
CODE-15	precision	0.654	0.610	-0.044
CODE-15	recall	0.741	0.856	0.115
CODE-15	f1	0.672	0.678	0.007
PTB-XL	accuracy	0.975	0.968	-0.007
PTB-XL	precision	0.575	0.538	-0.038
PTB-XL	recall	0.728	0.696	-0.032
PTB-XL	f1	0.614	0.575	-0.039

Table 12: Final aggregated metric differences. Positive values favor the first method in each comparison.

Comparison	Accuracy	Precision	Recall	F1
Individual cross-domain – in-domain	-0.070	-0.152	-0.049	-0.145
Balanced FL – in-domain	0.014	0.010	0.010	0.011
Balanced FL – foreign	0.084	0.162	0.060	0.156
Unbalanced low – individual	0.129	0.305	0.226	0.294
Unbalanced low – balanced	0.091	0.181	0.130	0.164
Unbalanced high – individual	-0.006	-0.042	0.012	-0.027

4 Main Answers to the Research Questions

RQ1: Does FL benefit scenarios with few local examples? Yes. With 1k examples per client, balanced FL improves macro F1 on all datasets. The gains are +0.212 for CODE-15, +0.135 for PTB-XL, and +0.041 for Chapman-Shaoxing relative to individual training. The improvement is also strong relative to foreign-model transfer, showing that FL produces a more useful shared representation than relying on models trained on other datasets.

RQ2: Does the gain persist with more local data? Partially. The local in-domain gain decreases with 5k and 10k examples. Individual models become strong enough to specialize to their own dataset, especially for PTB-XL and CODE-15. However, balanced FL remains better than the mean foreign model in every 5k and 10k target evaluation. Therefore, the persistent benefit of FL is not necessarily superior in-domain specialization, but improved collaborative generalization.

RQ3: Do larger clients help a low-resource client? Yes. In the unbalanced setting, the 1k client improves in every case. Macro F1 increases by +0.312 for CODE-15, +0.359 for PTB-XL, and +0.209 for Chapman-Shaoxing relative to individual 1k training. The unbalanced model also outperforms balanced 1k FL for the low-resource client, indicating that the additional examples from high-resource clients transfer useful information.

RQ4: Does collaboration harm high-resource clients? Slightly on average, but not uniformly. Across high-resource clients, unbalanced FL reduces macro F1 by -0.027 on average. CODE-15 as a high-resource client is most negatively affected, PTB-XL is modestly affected, and Chapman-Shaoxing improves. The cost is substantially smaller than the low-resource benefit, suggesting a favorable tradeoff when the goal is to support institutions with limited labeled ECGs.

5 Limitations

This study has several limitations. First, the clients are public datasets rather than real hospitals participating in a live FL deployment. Therefore, the experiment simulates institutional heterogeneity but does not cover operational issues such as network failures, institutional governance, authentication, auditability, or update security. Second, although FL avoids direct sharing of raw ECGs, the experiments do not implement formal privacy mechanisms such as secure aggregation, differential privacy, or update compression. In real clinical deployment, these would be important because model updates can still leak information under some threat models.

Third, the cross-domain results use thresholds selected on the target validation set. This is appropriate for studying target-calibrated transfer, but it does not measure calibration-free deployment in a new institution without labels. Fourth, computational constraints limited the explored sample-size regimes to 1k, 5k, and 10k examples per client. Larger regimes would be needed to determine when local training fully dominates FL or when FL continues to provide regularization. Fifth, the present experiments focus on a single FL algorithm, FedAvg-style aggregation. Other methods designed for statistical heterogeneity, such as proximal regularization, client-specific batch normalization, personalization layers, or mixture-of-experts approaches, may improve the tradeoff between low-resource benefit and high-resource cost.

Finally, the evaluation is based on six harmonized target abnormalities. This makes the cross-dataset comparison feasible, but it also discards dataset-specific label richness. PTB-XL and Chapman-Shaoxing contain broader diagnostic taxonomies, and CODE-based datasets contain clinically valuable annotations beyond the six targets considered here. Future work

should examine whether FL remains beneficial under larger label spaces and more realistic annotation mismatch.

6 Conclusion

This project evaluated federated learning for multilabel ECG abnormality classification using three heterogeneous public ECG datasets as simulated clients. The datasets differ strongly in positive-label density, class prevalence, multilabel burden, age distribution, and sex distribution. These differences create a realistic cross-silo challenge: each client represents a different empirical ECG domain.

The results show that FL is most useful in low-resource settings. With only 1k local examples per client, balanced FL improves all clients over individual training. As local data increase to 5k and 10k, individual in-domain models become more competitive and often outperform the global federated model locally, especially on PTB-XL and CODE-15. However, FL continues to improve over foreign-model transfer, showing that it remains useful for cross-domain robustness.

The strongest result is in the unbalanced setting. When one client has only 1k examples and the other two have 10k, the low-resource client improves substantially in every scenario. The average low-resource macro-F1 gain is +0.294 over individual training and +0.164 over balanced 1k FL. High-resource clients experience a small average macro-F1 reduction of -0.027. Therefore, the collaboration tradeoff is favorable when the goal is to support institutions with limited labeled ECG data.

Overall, the study supports federated learning as a promising strategy for collaborative ECG modeling under data scarcity and cross-dataset heterogeneity. It should not be interpreted as a universal replacement for strong local in-domain training, but rather as a mechanism for improving low-resource clients and producing models that generalize better across heterogeneous ECG populations.

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